

Genetic report for longevity

Ketosis and Ketone Bodies

Production and use of ketone bodies (HMGCS2, OXCT1, MPC1, SLC2A4, PPARA, MCT1), beta-hydroxybutyrate, acetoacetate and acetone scores, ketogenic diet, LDL and dapagliflozin

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The ketone bodies (beta-hydroxybutyrate, acetoacetate and acetone) are made in the liver from fat when there is little insulin: fasting, long exercise, ketogenic diet and, in the case of Sérgio F, also dapagliflozin (SGLT2 inhibitors lower insulin and rise the ketone bodies). Beta-hydroxybutyrate is fuel for the heart, muscle and brain and also acts as a signal (histone inhibits and NLRP3 inflammasome). This report reads the genes of production (HMGCS2, PPARA) and use (OXCT1, MCT1 transporter), the loci of large GWAS metabolites by nuclear magnetic resonance (UK Biobank) and three loci scores: beta-hydroxybutyrate, acetoacetate and acetone in circulation. Basal ketone levels are an informative trait: higher or lower is not, by itself, good or bad. The main practical point is safety: dapagliflozin with strict ketogenic diet, long fasting or vomiting increases the risk of ketoacidosis with normal glucose.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

11

variants

6

in the chip

5

imputed
(DR2 $\geq$ 0.8)

0

not found

3

Scores with
percentile

Overall reading. The ketone bodies scores are separated: acetoacetate at the 97th percentile and acetone in the 94 (acetone comes from acetoacetate, so both walk together), but beta-hydroxybutyrate in the 16. Beta-hydroxybutyrate and acetoacetate convert into one another, and the proportion between them depends on the redox state of the liver mitochondria; genetics points to a more displaced proportion for acetoacetate. All this is informative: it is not good or bad. In genes: **OXCT1 C/C** (2 copies of the allele that rises in circulation, the strongest locus), HMGCS2 C/C and SLC2A4 C/C (common alleles that rise them), MPC1 A/A and PPARA rs135549 C/C without the alleles that rise them; MCT1 A/T, PPARA L162V C/C and intron 7 C/G without judgment. For the diet and drugs: insulin in low genetic fasting (the 5th percentile) and HbA1c in 7, that is, it enters into ketosis with ease, which adds to the effect of dapagliflozin and retatrutide (come less). APOE ϵ 3/ ϵ 3 (without the exaggerated response of LDL to saturated fat from ϵ 4 carriers) and LDL polygenic at the 20th percentile. UGT1A9*3 absent (daily elimination of dapagliflozin). The most important practical point is safety: dapagliflozin with strict ketogenic diet, long fasting, vomiting or dehydration can give ketoacidosis with normal glucose.

Favourable

- **LDL (PGS000892)**: the 20th percentile (Lipid Report in Detail)
- **Fasting insulin (PGS000308)**: the 5th percentile (Glycosis, Insulin and Diabetes reports)
- **HbA1c (PGS001953)**: the 7th percentile (Glycosis, Insulin and Diabetes reports)
- **UGT1A9 (*3) rs72551330 TT**: UGT1A9*3 (M33T): reduced activity allele; increased exposure to dapagliflozin in pharmacokinetic studies

Unfavourable

None.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 6 chip genotypes, 0 needed complement, 1 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Scores of ketone bodies in circulation

Three loci scores of GWAS Catalog (beta-hydroxybutyrate, acetoacetate and acetone; especially the GWAS of metabolites in UK Biobank). Informative: a high percentile indicates a tendency for higher basal levels (in short fasting, without ketogenic diet).

Ageing mechanisms: Deregulation of Nutritional Sensing; Metabolic Homeostasis Decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Beta-hydroxybutyrate (GWAS loci)	No judgement (informative)	the 16th percentile · z=-1.01	Informative: higher score = basal beta-hydroxybutyrate. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 38 of 40 loci with genotype. Studies: Farashi S (3-Hydroxybutyrate levels); Richardson TG (3-Hydroxybutyrate levels); Yuan F (3-Hydroxybutyrate levels); Zoodsma M (3-Hydroxybutyrate levels). Variants with genotype: 38 of 40 (covering of 96.4% coefficients).
Acetoacetate (GWAS loci)	No judgement (informative)	the 97th percentile · z=2.02	Information: higher score = lower basal acetoacetate. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 18 of 19 loci with genotype. Studies: Zoodsma M (Acetoacetate levels). Variants with genotype: 18 of 19 (covering of 96.6% coefficients).
Acetone (GWAS loci)	No judgement (informative)	the 94th percentile · z=1.57	Information: higher score = higher basal acetone. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 25 of 29 loci with genotype. Studies: Farashi S (Acetone levels); Karjalainen MK (Acetone levels); Richardson TG (Acetone levels); Zoodsma M (Acetone levels). Variants with genotype: 25 of 29 (covering of 75.6% coefficients).

Favourable

None.

Unfavourable

None.

Production and use: HMGCS2, OXCT1, MPC1, SLC2A4, PPARA and MCT1

HMGCS2 (restricting step of ketogenesis in the liver, linked by PPARα), OXCT1 (SCOT, the enzyme that allows tissues to use ketone bodies; the liver does not have it), MPC1 (transporter from pyruvate to mitochondria), SLC2A4 (GLUT4, glucose uptake by muscle), PPARA (fast transcription factor) and SLC16A1 (MCT1, transports ketone bodies and lactate). In the tables, the "favourable allele" of the GWAS loci is here the allele that rises the ketone bodies in circulation; it remains without judgment. Severe deficits (HMGCS2, SCOT, ACAT1, HMGCL) are rare and do not read by chip.

Ageing mechanisms: Mitochondrial dysfunction; Deregulation of Nutritional Sensing; Insulin resistance

Favourable	Unfavourable
None.	None.

Non-judgmental variants (informative) (8)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SLC16A1 (MCT1)	rs1049434 1:112913924 · A/G/T	AT	chip	—	weak	MCT1 Asp490Glu: T allele was associated with lower lactate transport and ketone bodies into the cells, with small effect; mixed replication.
PPARA	rs1800206 22:46218377 · C/G	CC	chip	—	weak	PPARA L162V: allele V (G) alters the response to lipids and fibrates; uncommon (~5%).
PPARA (7 intron)	rs4253778 22:46234737 · G/C/T	CG	chip	—	weak	PPARA intron 7 G/C: C allele reduces the expression of PPARα in the muscle (more use of glucose, less fat); associated with the power profile in the studies of athletes.
OXCT1	rs1533014 5:41849497 · T/C	CC	imputation DR2=1.00	C (2; 31%)	moderate	OXCT1 (SCOT): the strongest locus for circulating ketone bodies; variants that lower peripheral use rise blood levels. Meaning: GWAS Catalog: Acetone levels (Zoodsma M 2025; p=1e-169); C ↑ allele
HMGCS2	rs2582783 1:119782372 · T/A/C/G	CC	chip	C (2; 71%)	moderate	HMGCS2: mitochondrial synthesis of HMG-CoA in the liver, a limiting step of ketogenesis. Meaning: GWAS Catalog: Acetone levels (Zoodsma M 2025; p=1e-24); C ↑ allele
MPC1	rs117351628 6:166353253 · A/G	AA	imputation DR2=0.94	G (0; 2%)	moderate	MPC1: transporter of mitochondrial pyruvate; less entry of pyruvate diverts metabolism to fat and ketone bodies. Meaning: GWAS Catalog: Acetoacetate levels (Zoodsma M 2025; p=1e-83); G ↑ allele
SLC2A4	rs117643180 17:7282460 · C/A/T	CC	imputation DR2=0.99	C (2; 97%)	moderate	SLC2A4 (GLUT4): glucose uptake by muscle and fat, stimulated by insulin and exercise. Meaning: GWAS Catalog: Acetone levels (Richardson TG 2022; p=4e-15); allele C ↑
PPARA	rs135549 22:46157413 · T/C	CC	chip	T (0; 61%)	moderate	PPARA: transcription factor of fasting; binds beta-oxidation and HMGCS2. Meaning: GWAS Catalog: Acetone levels (Zoodsma M 2025; p=1e-10); T ↑ allele

Ketogenic diet: LDL, insulin and drugs

The ketogenic diet rises the LDL in part of people, sometimes much (more saturated fat and in lean people and sensitive to insulin). APOE $\epsilon 4$ increases this response; $\epsilon 3/\epsilon 3$ does not. Dapagliflozin is eliminated by glucuronidation (UGT1A9). The LDL scores, triglycerides and insulin already calculated in other reports are added.

Ageing mechanisms: Vascular Aging; Insulin resistance; Hyperglycemia and Intolerance to Glucose

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
LDL (PGS000892)	Favourable	the 20th percentile	Genetic basis of low LDL (typical label in the origin report, at the limit of favourable); does not predict individual response to ketogenic diet. Source: Lipid Report in Detail.
Triglycerides (PGS002719)	Typical / intermediate	the 68th percentile	"typical" label. Source: Lipid Report in Detail.
Fasting insulin (PGS000308)	Favourable	the 5th percentile	Genetically low basal insulin: it enters ketosis more easily. Source: Glucose, Insulin and Diabetes report.
HbA1c (PGS001953)	Favourable	the 7th percentile	"Favourable" label. Source: Glucose, Insulin and Diabetes report.
2 type diabetes (PGS000014)	Typical / intermediate	the 57th percentile	"typical" label. Source: Glucose, Insulin and Diabetes report.

Favourable

- LDL (PGS000892) (the 20th percentile)
- Fasting insulin (PGS000308) (the 5th percentile)
- HbA1c (PGS001953) (the 7th percentile)
- UGT1A9 (*3) rs72551330 TT

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
UGT1A9 (*3)	rs72551330 2:233672032 T/A/C	TT	chip	C (0; 2%)	moderate	UGT1A9*3 (M33T): reduced activity allele; increased exposure to dapagliflozin in pharmacokinetic studies. Rare.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
APOE	rs429358 19:44908684 T/C	TT	imputation DR2=0.99	—	strong	APOE rs429358 T/T = without $\epsilon 4$ (with rs7412, $\epsilon 3/\epsilon 3$, as in the profile). The $\epsilon 4$ carriers increase the LDL with saturated fat; the $\epsilon 3/\epsilon 3$ has an average response. In GWAS, $\epsilon 4$ also touches beta-hydroxybutyrate.
UGT1A9	rs6714486 2:233671659 T/A	TT	imputation DR2=1.00	—	weak	UGT1A9 -275T>A: allele A was associated with more expression of UGT1A9 (the enzyme that eliminates dapagliflozin); clinical effect not demonstrated.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Metabolic Homeostasis Decline	—	—	—
Nutrient Sensing Deregulation	—	—	—
Mitochondrial dysfunction	—	—	—
Vascular Aging	—	LDL (PGS000892) (the 20th percentile); Fasting insulin (PGS000308) (the 5th percentile); HbA1c (PGS001953) (the 7th percentile); UGT1A9 (*3) TT	Triglycerides (PGS002719) (the 68th percentile); 2 (PGS000014) type diabetes (the 57th percentile)
Hyperglycemia and Intolerance to Glucose	—	LDL (PGS000892) (the 20th percentile); Fasting insulin (PGS000308) (the 5th percentile); HbA1c (PGS001953) (the 7th percentile); UGT1A9 (*3) TT	Triglycerides (PGS002719) (the 68th percentile); 2 (PGS000014) type diabetes (the 57th percentile)
Insulin resistance	—	LDL (PGS000892) (the 20th percentile); Fasting insulin (PGS000308) (the 5th percentile); HbA1c (PGS001953) (the 7th percentile); UGT1A9 (*3) TT	Triglycerides (PGS002719) (the 68th percentile); 2 (PGS000014) type diabetes (the 57th percentile)

Practical part

Ideas about ketosis, crossed with the genetic plan v2 and current medication (dapagliflozin, retatrutide, lithium 5 mg, telmisartan, rapamycin, tadalafil, low dose naltrexone, isotretinoin). The current stack remains as it is; everything is suggestion (not applied).

1. No strict ketogenic diet or long fasts with dapagliflozin

Genetics: Fasted insulin p5; acetoacetate p97; dapagliflozin

With a SGLT2 inhibitor, the strong restriction of carbohydrates (below ~50 g/day) or fasts of more than 24 hours increase the risk of ketoacidosis with normal glucose. If you want to try, only with the doctor, that you can decide to pause dapagliflozin before.

V2 genetic plan and medication: Compatible with the plan (without ketogenic diet in the plan). • *suggestion (not applied)*

2. Days of disease: vomiting, without eating or dehydration

Genetics: Dapagliflozin + retatrutide (nauseas); lithium

On days with vomiting or unable to eat, talk to the doctor about pause dapagliflozin on these days, as the plan already does with lithium. Drinking water and replenishing salt; dehydration also rises lithium.

V2 genetic plan and medication: Compatible with the plan (leisure paused in days of vomiting). • *suggestion (not applied)*

3. Having a ketone meter in the blood

Genetics: Dapagliflozin; genetic trend for high acetoacetate

Nausea, vomiting, abdominal pain, tiredness out of normal or fast breathing with dapagliflozin: measuring beta-hydroxybutyrate in the blood (the urine strips read acetoacetate and deceive). Values above ~1.5 mmol/L with symptoms require medical contact on their own day.

V2 genetic plan and medication: No change in the plan. • *suggestion (not applied)*

4. To reduce carbohydrates, unsaturated fat

Genetics: APOE ε3/ε3; LDL p20

In a lower diet in carbohydrates (non-ketogenic), prefer olive oil, dried fruits, avocado and fish to butter and fat meat, and keep the fiber high. Repeat LDL and ApoB 6–8 weeks after any major change: the lower, the better.

V2 genetic plan and medication: Compatible with the plan (LDL/ApoB: the lower, the better; the more fiber). • *suggestion (not applied)*

Avoiding or with caution

- Strict ketogenic diet and fasts of more than 24 hours with dapagliflozin, without medical follow-up. *suggestion (not applied)*
- Exogenous ketones (esters or salts of beta-hydroxybutyrate) and high dose MCT oil added to dapagliflozin. *suggestion (not applied)*
- Dehydration (dapagliflozin + ketosis + lithium). *suggestion (not applied)*

Limitations

- The scores measure basal levels of ketone bodies (samples without long fasting, in a population without ketogenic diet); they do not measure the ability to enter into ketosis or the response to diet, fasting or drugs.
- The GWAS of nuclear magnetic resonance metabolites have many loci shared with lipids and glucose (APOE, GCKR, TRIB1); part of the score reflects these traits.
- The severe hereditary deficits of ketogenesis and ketolysis (HMGCS2, OXCT1, ACAT1, HMGCL) and CPT1A P479L do not read with confidence per chip.
- The percentiles compare with a reference population of 633 genomes.
- Only genes do not include ketones, glucose or lipids. It is neither a diagnosis nor a prescription.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: .

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Karjalainen MK et al. Genome-wide characterization of circulating metabolic biomarkers (Nature 2024).
- Richardson TG et al. Characterizing metabolomic signatures of lipid-modifying therapies through drug target mendelian randomization (PLoS Biol 2022).
- Zoodsma M et al.: GWAS of nuclear magnetic resonance metabolites in UK Biobank (GWAS Catalog entries).
- Newman JC, Verdin E. Ketone bodies as signaling metabolites (Trends Endocrinol Metab 2014).
- Puchalska P, Baxter PA. Multi-dimensional roles of ketone bodies in fuel metabolism, signaling, and therapeutics (Cell Metab 2017).
- FDA Drug Safety Communication: SGLT2 inhibitors and ketoacidosis (2015, later updated).
- Norwitz NG et al. Elevated LDL cholesterol with a carbohydrate-restricted diet: evidence for a "lean mass hyper-responder" phenotype (Curr Dev Nutr 2022).
- GWAS Catalog (NHGRI-EBI), full download of associations.

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